

ENVIRONMENTAL SECURITY TECHNOLOGY CERTIFICATION PROGRAM (ESTCP) BROAD AGENCY ANNOUNCEMENT FOR ENVIRONMENTAL AND INSTALLATION ENERGY DEMONSTRATION AND VALIDATION

Reference: Broad Agency Announcement (BAA), 22 August 2025
U.S. Army Corps of Engineers, Humphreys Engineer Center Support Activity

1. INTRODUCTION

The Environmental Security Technology Certification Program (ESTCP) is the Department of Defense's (DoD) demonstration and validation program for innovative technologies that enhance military readiness, improve installation resilience, and strengthen America's defense infrastructure. The program focuses on deploying common sense, cost-effective, and scalable solutions that ensure operational effectiveness, improve warfighter capabilities, and reduce regulatory burdens.

ESTCP supports the demonstration and validation of technologies that:

- a) Increase energy security and grid independence through enhanced domestic energy production, microgrids, and advanced power generation
- b) Improve operational efficiency by reducing sustainment costs, enhancing logistics, and ensuring mission continuity
- c) Enhance installation resilience against extreme weather events, cyber threats, and physical disruptions
- d) Optimize defense infrastructure to ensure military bases, training grounds, and facilities remain fully functional and cost-efficient.

The program gives priority to technologies that strengthen the American industrial capacity, reduce reliance on foreign supply chains, and align with U.S. national security objectives. Technologies must be sufficiently mature for field demonstration, with proven feasibility and the potential for rapid transition into widespread DoD use. ESTCP does not fund regulatory compliance studies but instead supports actionable technology development that directly contributes to mission assurance and warfighter readiness.

1.1 GENERAL INFORMATION FOR PROPOSERS

ESTCP seeks white papers from private sector organizations proposing innovative, cost-effective solutions to address DoD's most pressing energy security, infrastructure resilience, and sustainment challenges. Technologies should have the potential to significantly enhance mission assurance, reduce logistical vulnerabilities, and increase operational efficiency. Topic areas defined by this Broad Agency Announcement (BAA) are located in Appendix A of this document. This BAA is for Private Sector Organizations.

White papers should focus on technologies that directly enhance DoD capabilities rather than broad environmental research.

Awardees under this BAA will be selected through a multi-stage review process, including a brief white paper and a full proposal. Some proposals may require an oral presentation to the ESTCP Technical Committee (ETC). To be eligible for consideration, proposers must submit a white paper. Any white paper submitted shall be in response to only one of the topic areas set forth in this announcement. White papers marked with ITAR, EAR, a limited distribution statement, Controlled Unclassified Information (CUI), or as company proprietary are not accepted. Please note, only government employees or Department of Defense contractors who have signed non-disclosure agreements have access to ESTCP proposals. Proposers may respond to more than one topic area

with separate white papers.

After evaluation of white papers, ESTCP will contact all proposers and either request or not request each to submit a full proposal. At that time, ESTCP will provide detailed instructions for the full proposal format. Full proposals may not be submitted outside the white paper process. Any full proposal that has not been reviewed in the white paper phase will not be evaluated nor considered for award under this BAA. Due to the volume of white papers anticipated, ESTCP will not provide debriefs on those that are not requested to submit a full proposal.

Based on an evaluation of the written full proposal, and pending recommendation by the ETC, ESTCP will notify each proposer as to whether the Government Contracting Officer wishes to enter into negotiation for the award of a contract. Proposers are advised that only the Contracting Officer is legally authorized to bind the Government. ESTCP reserves the right to recommend for award any, all, or none of the proposals received. ESTCP also reserves the right to recommend a portion of the work proposed in any single proposal for award. There is no commitment by ESTCP to make any recommendations for contract awards, nor to be responsible for any money expended by the proposer before contract award is made. It is the sole responsibility of the proposer to make certain the proposal is properly received by ESTCP.

The ESTCP Office manages the BAA solicitation along with the U.S. Army Corps of Engineers, Humphreys Engineer Center Support Activity (HECSA) Contracting Center in Alexandria, Virginia. For contracting or small business information regarding this BAA solicitation, contact Shannon Benson: 703-428-7407, Shannon.N.Benson@usace.army.mil or the HECSA Small Business Representative, Monique Holmes: 202-934-1816, HECSASmallBusinessProgram@usace.army.mil. General SERDP procedural questions may be referred to the SERDP office at 571-372-6565. For technical information regarding a topic area, contact the individual listed in Appendix A.

1.2 EVALUATION SCHEDULE

This BAA is open for one year from the date of release or until it is replaced by a subsequent announcement, whichever comes first.

White papers will not be evaluated against each other since they are not submitted in accordance with a common work statement. ESTCP's intent is to review white papers as soon as possible after they are submitted; however, white papers may be reviewed periodically for administrative reasons. Proposers will be notified on the decision regarding their white paper as soon as possible following review.

Full proposals will be evaluated as soon as possible after they are submitted. Review of full proposals may require input from external peer reviewers or other technical experts. Proposals may be required to brief the ETC depending on their size and scope. ETC meetings are typically held three times each fiscal year.

Contracts are anticipated to be awarded three to six months following final review and approval of each proposal, depending on availability of funding.

2. WHITE PAPER INSTRUCTIONS

2.1 WHITE PAPER LENGTH AND FORMAT

White papers shall be no longer than two (2) pages, type face not less than 11-point, and margins not less than one inch on all sides. All white papers shall be submitted as a single PDF file containing all sections outlined below.

2.2 WHITE PAPER CONTENT

White papers must describe the merits and objectives of the proposer's project in response to the respective topic area. The white paper should concisely describe the following:

1. **Proposal Title**
2. **Lead Principal Investigator**
3. **Lead Organization**
4. **Objectives:** State the proposed objectives and how the project is responsive to the objectives articulated in the topic area.
5. **Technical Approach:** Structure the technical approach in terms of tasks to be accomplished in accordance with guidance provided in the individual topic areas. Clearly define the performance objectives for the technology, including their associated performance metrics, data requirements, and success criteria, and provide a table containing this information. Refer to Table 1 in the [Demonstration Plan Guidance](#) for the type of information expected. Individuals submitting to an Installation Energy and Water topic should refer to Section 5.0 of the ESTCP Installation Energy and Water Demonstration Plan Guidance for more perspective on test design.
6. **Cost Estimate:** The estimated total costs by year. A detailed breakout of costs is not required or desired in the white paper. A detailed cost breakout will be provided in the full proposal, if requested. If selected for funding, proposers will be required to provide a certificate of current cost or pricing data prior to award in accordance with Federal Acquisition Regulation (FAR) 15.403-4(a)(1) if the total contract value is expected to exceed \$2,000,000.00 to the cognizant contracting office.
7. **Literature Citations:** Provide literature citations for any material cited.

3. SUBMITTAL

White papers are submitted via the SERDP and ESTCP Management System (SEMS). No electronic mail, faxed, or hard copy white papers will be accepted.

Complete all steps below in order to submit a white paper.

- Create a single PDF file that contains all required sections outlined in the white paper guidance.
- Click the link below that corresponds to the Topic the white paper responds to (create a SEMS account if needed):
 - [Environmental Restoration](#)
 - [Installation Energy and Water](#)
 - [Munitions Response](#)
 - [Natural Hazards Adaption](#)
 - [Mission-Driven Landscape Approaches to Regions Critical to DoD Operations](#)
 - [Resource Conservation and Resilience](#)
 - [Weapons Systems and Platforms](#)

Enter all required white paper details into SEMS. If the white paper is associated with an existing ESTCP project, enter the project number on the first page of the white paper submission wizard in SEMS. White paper details may be saved and edited prior to final submission.

- Before submitting, add the system-generated proposal number (displayed in front of the white paper title on the “Manage Proposals” page), project title, lead PI name and organization to first page of white paper as instructed in Section 2.2 above.
- Upload the final white paper file.
- Submit the white paper. Only white papers that have been submitted will be considered. White papers with a “Saved” status will not be reviewed.

Edits cannot be made once the white paper has been submitted. SEMS will display an on-line confirmation message and send an email notification to the proposer. It is the sole responsibility of the proposer to make certain the white paper is properly received by ESTCP. For questions, contact the ESTCP Office at 571-372-6565.

Note: A signed cover page is **not required** for white papers.

4. EVALUATION FACTORS FOR WHITE PAPERS AND FULL PROPOSALS

The following evaluation factors will be the sole basis for reviewing white papers and full proposals submitted in response to this BAA. ESTCP relevance is a pass/fail criteria evaluated at the white paper stage only; white papers not passing this gate will not be further evaluated. Among the other evaluation factors for both white papers and full proposals, Technical Merit is more important than Cost/Benefit of Technology, Transition Potential, and Cost of Proposal.

ESTCP RELEVANCE (PASS/FAIL)

An assessment will be made whether the white paper submission responds to the DoD requirement as described in the topic area. ESTCP Relevance is a threshold review and if the white paper is determined not to be relevant, no further evaluation of the white paper will be made.

TECHNICAL MERIT

An assessment of the technical merit of the white paper will be made. Factors to be considered include: (a) the methodology is scientifically sound; (b) the technology is innovative and is the current state-of-the-art, (c) the technical risks are well characterized, and (d) the technical team is qualified to execute the proposed project (full proposals only).

COST/BENEFIT OF TECHNOLOGY (FULL PROPOSALS ONLY)

An assessment of the cost/benefit of the proposed technology, if it were deployed, will be made. Factors to be considered include: (a) the projected cost savings and/or risk reduction are significant; (b) the projected benefits are reasonable and consistent with the proposed technology; and (c) the payoffs from the proposed technology are commensurate with the projected costs and risks.

TRANSITION POTENTIAL (FULL PROPOSALS ONLY)

An assessment of the potential for a successful transfer of the technology to the DoD user will be made. Factors to be considered include: (a) there is a well-defined DoD user for the technology; (b) there are clearly identified activities that will support and enhance the transfer of the technology; and (c) the technology can be implemented within DoD.

COST OF PROPOSAL (FULL PROPOSALS ONLY)

Cost realism and reasonableness may be considered during the selection of acceptable proposals in accordance with FAR 35.016(e). Costs should be appropriate and traceable to the level of effort required to execute the project.

5. APPENDIX A: ESTCP TOPIC AREAS

ENVIRONMENTAL RESTORATION

ESTCP's Environmental Restoration Program Area focuses on restoring and managing impacted lands on current and former military installations while ensuring cost-effective and mission-ready solutions. The program supports the demonstration of innovative technologies to assess, remediate, and manage chemicals of concern in soil, sediments, and water sources. The priority is on practical, field-ready approaches that enhance operational flexibility and reduce long-term costs and obligations for the DoD.

Characterization and Monitoring

Demonstration of methodologies or technologies that improve the efficiency and accuracy of long-term monitoring for chemicals of concern in soils, sediments, and water. Solutions should emphasize rapid deployment, cost savings, and operational readiness.

Reduction in Cost to Complete

Innovative tools and technologies are sought to streamline performance assessments and optimize treatment for impacted groundwater or aquatic sediments. The focus is on reducing costs and timelines associated with remediation while ensuring compliance with DoD mission requirements. Sites impacted by PFAS, chlorinated solvents, munitions constituents, PCBs, and PAHs are of primary concern.

Reduce Source Loading of Munitions Constituents

Development and demonstration of technologies that reduce source loading of munitions constituents during routine DoD operations and demilitarization activities. The goal is to enhance sustainability and operational efficiency while minimizing environmental impact.

Stormwater Treatment

Cost-effective and low-maintenance solutions for management and treatment of stormwater runoff from DoD facilities. Technologies should integrate with existing Best Management Practices and support compliance with National Pollution Discharge Elimination System (NPDES) permit requirements while minimizing regulatory burden.

Wastewater Treatment

Demonstration of decentralized wastewater treatment and recycling systems that are energy-efficient, low-maintenance, and capable of supporting fixed installations. Solutions that operate in an energy-neutral configuration or produce materials that can be converted into power are of interest.

Risk Assessment

Technologies that improve the military's ability to assess and predict human and ecological risks from contaminants of concern. Areas of focus include exposure pathways to humans, ecological receptor impacts, and streamlined tools for assessing environmental risks.

To ensure alignment with DoD priorities, SERDP and ESTCP periodically conduct workshops to identify and prioritize demonstration and validation needs. Proposers are encouraged to refer to

the following report for additional insight:

- [Summary Report](#) from the November 2022 SERDP and ESTCP *Strategic Workshop on Management of PFAS in the Environment*

Data needs identified during these workshops are relevant to several of the specific areas of interest listed above.

Point of Contact

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INSTALLATION ENERGY AND WATER

ESTCP's Installation Energy and Water (EW) program area supports the demonstration and validation of innovative technologies that have strong potential to cost-effectively improve the resilience of DoD installations. This program prioritizes cost-effective, scalable, and cyber-secure solutions that reduce operational costs while ensuring the reliability of energy and water systems critical to national defense. Key areas of interest include strengthening installation resilience, optimizing infrastructure efficiency, and integrating advanced energy and water management solutions that support DoD's readiness objectives.

Secure and Reliable Energy/Water

Innovative technologies and solutions are sought to improve the security, reliability, and flexibility of energy and water systems that sustain mission-critical operations. DoD installations vary in size and complexity, ranging from small facilities to expansive installations covering over 1 million acres. This program seeks innovative, low-cost solutions that enhance infrastructure resilience, protect against disruptions, and integrate seamlessly with existing systems. Proposals should focus on technologies that enhance the security, reliability, and efficiency of existing energy and water systems, reduce vulnerabilities by diversifying energy sources and strengthening cyber resilience, improve interoperability to lower design and integration costs, and maximize the use of existing on-site distributed energy resources (DER) to enhance operational flexibility.

Infrastructure and Facility Energy Reliability

DoD manages over 300,000 buildings across approximately 500 installations nationwide. To maintain mission effectiveness and reduce logistical burdens, the program seeks innovative, cost-effective, durable, and low-maintenance solutions. Technologies under consideration include generation, storage, and backup systems that ensure uninterrupted energy supply, hardening of infrastructure against external threats and system failures, practical, field-ready water management solutions that improve operational sustainability, and integrated energy and water management systems to support operational efficiency and cost control.

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MUNITIONS RESPONSE

ESTCP's Munitions Response program area supports the demonstration of innovative technologies that can characterize, remediate, and manage sites affected by military munitions with a focus on cost-effective and scalable solutions. Ensuring national security while maintaining environmental and public safety is a priority. The program seeks technologies that enhance operational efficiency, reduce remediation costs, and improve detection and classification capabilities.

Many affected sites contain munitions in underwater environments, with depths typically less than 5 meters, though concerns extend to depths of up to 35 meters. These environments include ponds, lakes, rivers, estuaries, and coastal or open ocean areas. Munitions of interest range from small projectiles and mortars to large bombs.

Current research priorities include:

Wide Area and/or Detailed Survey Techniques

Technologies to conduct large-scale, cost-effective surveys to identify munitions concentrations and clear areas are of interest. Sensor modalities should maximize coverage rates while maintaining reliable probabilities of detection and classification. Areas identified as contaminated require detailed data collection to assess contamination levels and define appropriate remediation strategies. Detection and classification technologies should provide high accuracy and precise location tracking to facilitate retrieval or continued monitoring of munitions.

Cost-Effective Recovery and Disposal Methods

Improved recovery and disposal techniques are sought to efficiently and safely remove munitions from the underwater environment. Current practices relying on divers for manual retrieval are costly and hazardous. The program seeks innovative recovery technologies suited for shallow water environments, particularly where munitions may pose a risk to the public, such as areas frequented by recreational divers. Proposed methods must address explosive safety concerns and provide cost-effective, safe, and environmentally responsible remediation solutions. Technologies that reduce reliance on blow-in-place operations to mitigate impacts on marine life are of particular interest.

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NATURAL HAZARDS ADAPTATION

The Natural Hazards Adaptation (NH) program area seeks cost-effective, mission-driven solutions to strengthen Department of Defense (DoD) infrastructure against natural hazards while ensuring military readiness and operational superiority. The focus is on practical, resilient, and fiscally responsible approaches that reduce exposure or improve recovery from natural hazard impacts without increasing regulatory burden. This program area prioritizes risk-based investments and dual-purpose technologies that enhance DoD capabilities and national security in key DoD geographies while minimizing costs and maximizing operational effectiveness. Projects should align with mission-critical infrastructure protection, address multiple hazards, improve military installation planning or preparedness, or advance installation energy resilience.

Hazard Adaptation Technologies

High-impact, low-cost adaptation technologies are sought to protect installations from extreme weather events, wildfire, recurrent flooding, coastal erosion, permafrost thawing, drought, and saltwater intrusion. Solutions should emphasize durability, long-term cost savings, and operational efficiency. Priority areas include military-grade infrastructure hardening, reinforced building materials, storm protection barriers, and enhanced water management systems. Technologies that reduce maintenance costs and enhance installation or mission resilience will be prioritized.

Data Collection in Support of Engineering Design and Construction

To ensure mission effectiveness, DoD installations require accurate and actionable data to support infrastructure design, force protection, and base resilience in key DoD geographies, defined as areas of high concentration of DoD installations and assets, such as the arid Southwest, Southeast Coastal Plain, Alaska, and Pacific Islands. Proposals should focus on efficient, field-ready data collection techniques, including geospatial intelligence, military-focused GIS applications, and real-time environmental risk monitoring. Data should be structured to streamline military planning, optimize force deployment, and reduce infrastructure costs. Solutions should integrate with targeted DoD end-user communities and existing DoD installation management systems to ensure ease of adoption.

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MISSION-DRIVEN LANDSCAPE APPROACHES TO REGIONS CRITICAL TO DOD OPERATIONS

The DoD manages over 27 million acres of land essential to military operations, training, and national security. Ensuring the resilience of these installations is critical to sustaining military readiness and maintaining global force projection. Strategically significant regions or key DoD geographies, such as Alaska, Hawaii, the Southern Border, the southeastern U.S., Southern California, the eastern seaboard, and the southwestern U.S., face unique operational challenges that could affect military effectiveness. These areas require proactive risk management and mission-focused innovation to prevent disruptions to key defense infrastructure and training capabilities. DoD investments must prioritize installation security, adaptability, and long-term mission sustainability, ensuring that solutions integrate seamlessly into warfighter readiness, logistics efficiency, and force resilience.

The Mission-Driven Landscape Approaches topic area seeks innovative, cost-effective, and operationally relevant strategies that enhance the resilience of DoD infrastructure and training environments. Proposals should focus on strategic military advantage, resource efficiency, and installation readiness. Projects must align with DoD priorities, ensuring that mission-critical infrastructure remains functional and that investments maximize national security benefits.

Proposals should address several of the following key requirements:

- Define the strategic importance of the DoD geography, identifying current and potential risks to military operations and infrastructure.
- Demonstrate operational benefits of proposed tools, strategies, and technologies by accelerating infrastructure resilience to meet national defense priorities.
- Engage military end users to ensure alignment with DoD mission objectives and operational requirements.
- Leverage dual-purpose technologies to advance installation resilience to natural hazards.
- Ensure scalable implementation by defining clear pathways for adoption across multiple DoD installations, with measurable success indicators and impact assessments.
- Provide a clear return on investment (ROI), demonstrating cost-effectiveness and mission benefits for both near-term and long-term applications.
- Focus on mission-critical decision support, utilizing existing, geographically relevant environmental data to inform strategic planning without redundant climate assessments.
- Support rapid deployment of proven technologies, prioritizing efficiency and practicality within a budget of <\$500K and an 18–24-month timeline for execution.

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RESOURCE CONSERVATION

ESTCP's Resource Conservation and Resilience (RC) program area enhances DoD mission readiness by supporting the demonstration and evaluation of technologies that improve operational effectiveness and natural infrastructure stability. This program prioritizes cost-effective, mission-driven solutions that maintain and improve DoD training and testing capabilities while ensuring long-term resource reliability. Efforts focus on advancing technologies that support military operations, sustain training ranges, and optimize installation management to enhance national defense priorities.

Managing Prescribed Fires

Technologies are sought to improve the management of prescribed fires on and around DoD installations to ensure training and testing activities remain uninterrupted while preventing impacts on surrounding communities. Solutions should focus on tools that improve safe and effective use of prescribed fire, reduce the risk of uncontrolled wildfire, and minimize smoke exposure to nearby populations. Research should prioritize practical, field-ready approaches that improve fire management without increasing operational costs or logistical burdens.

Natural Resources

Natural resource technologies are sought to support the management of land, water, and species that are essential to DoD installations and their operational readiness. Solutions should ensure that training and testing conditions remain realistic and unimpeded while supporting resource management that enhances long-term operational effectiveness. Key areas of interest include technologies that monitor land use change, optimize land management for military use, enhance long-term range viability, and manage at-risk or invasive species that may impact future operations. Research should focus on scalable, low-maintenance solutions that improve land stability without adding unnecessary costs or administrative burdens.

Cultural Resource Technologies

To improve efficiency of compliance efforts, advanced technologies are sought that improve the efficiency and accuracy of cultural resource surveys and site mapping while minimizing costs or delays of legal requirements. Solutions should enhance the ability to quickly identify, document, and manage cultural resources using remote sensing, artificial intelligence, and uncrewed aerial systems. Technologies that accelerate compliance processes and improve data accuracy without delaying mission-critical projects are of interest.

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WEAPONS SYSTEMS AND PLATFORMS

ESTCP's Weapons Systems and Platforms program area supports the demonstration of innovative technologies and materials to ensure America's warfighters have access to current critical materials, next-generation materials, and manufacturing and repair processes to maximize the lethality and readiness of our weapons systems and platforms from production through decommissioning.

Manufacturing and Maintenance

Demonstration and validation of technologies that enable cost-effective maintenance and modernization of military platforms is of interest. Areas of focus include advanced coatings, primers, and surface treatments that improve durability and reduce corrosion, as well as alternative solvents and technologies for efficient material processing. Additional priorities include high-performance materials for additive manufacturing and advanced joining techniques to enhance structural integrity. Solutions should emphasize increased longevity and reduced sustainment costs for DoD assets while minimizing the need for costly protective measures such as extensive PPE and engineering controls. Demonstration and validation of non-flammable refrigerants and novel refrigeration technologies that do not use refrigerants are of significant interest to ground vehicles. The use of alternative flame retardants for soldier, vehicle, and support equipment applications ready for demonstration are of interest. Alternate chemical treatments and materials for soldier uniforms and equipment with commercially viable materials and processes are of interest. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposed technologies should have considerable laboratory results to indicate potential for successful demonstration/validation.

To ensure alignment with DoD priorities, SERDP and ESTCP periodically conduct workshops to identify and prioritize demonstration and validation needs. Proposers are encouraged to refer to the [ASETSDefense workshop](#) reports for additional insight.

Energetic Materials and Formulations

The demonstration and validation of alternative materials or fabrication methods for military propellants, explosives, and pyrotechnics are sought. Alternative materials or methods should improve performance, enhance safety, and reduce costs associated with operational requirements and waste disposal. Also of interest are innovative demilitarization techniques that improve disposal efficiency without compromising mission capability. Solutions should streamline disposal processes, lower costs associated with waste byproducts handling and ensure compliance with DoD operational requirements with minimal regulatory burden. Demonstration and validation of prototype demilitarization units are sought. Technologies to rapidly, safely, and effectively remove energetics from munitions and improve their processing for downstream destruction processes are also sought. Demilitarization technology should be applicable to conventional ammunition, rockets, explosives, or pyrotechnic devices and include demilitarization potential for propellants, explosives, plastic bonded-explosives, cast cure rocket propellants, or pyrotechnics. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposed technologies should have considerable laboratory results to indicate potential for successful demonstration/validation.

Additional information on our current and past efforts in energetic materials is available on our website.

Firefighting

Demonstration and validation of advanced fluorine-free firefighting foams (F3) that can extinguish class B fuel fires across multiple scenarios, particularly those involving Department of Defense (DoD) vessels and marine locations are sought. Technologies should be validated against the fire suppression criteria and other requirements as outlined in MIL-PRF-32725 while using saltwater dilution until a new fluorine-free specification is created for shipboard use. Alternative technologies in accordance with MIL-PRF-32725 for land-based use should demonstrate improved fire extinguishing performance, reduced cost, or reduced health hazard assessment relative to current products available on the qualified products database for MIL-PRF-32725. Alternative technologies are also sought to enable fire suppression within the cabin of vehicles, ships, and aircraft. Alternative fire suppressants and dispersion/distribution technologies will be considered. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposed technologies should have considerable laboratory results to indicate potential for successful demonstration/validation.

Critical Minerals, Chemicals, and Materials

Various critical minerals, chemicals, and materials are used on weapons systems and platforms, including, but not limited to neodymium, samarium, graphite, and carbon fiber. Technologies are sought to recover critical minerals and materials from DoD materiel that is not currently recycled and to identify improved recycling technologies and processes to increase recovery efficiency. Demonstration and validation of extraction and refining technologies with increased effectiveness are of interest. Demonstration and validation of alternatives to critical minerals, materials, and chemicals, including those restricted by TSCA regulations, for application to DoD materiel to improve resilience are sought. Proposed technologies should minimize or eliminate the use of TSCA-restricted chemicals, hazardous chemicals, and emerging chemicals of concern. Proposed technologies should have considerable laboratory results to indicate potential for successful demonstration/validation.

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